

Artificial Intelligence in the Knowledge Economy

Autonomy, capability, and a Lean verification exercise

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github.com/gsaco/ai-05-ide

Question and source

Who gains when knowledge becomes scalable?

We track bottom wages, top wages, and total output.

- Course reading: May 20, 2025, 39 pages.
- Lean source: arXiv v11, February 25, 2025, 35 pages.
- Focus: Propositions 5 and 6, then a three-type example.

Autonomy and capability are different dimensions

Autonomy: permitted roles

Autonomous AI can produce, work, and advise.

Co-pilot-only AI can advise.

Capability: knowledge

$$a = z_{AI} \in [0, 1).$$

The fraction of problems AI can solve.

Why the distinction matters

Autonomy changes outside options; capability changes which humans AI complements or substitutes for.

The agent's problem: organize knowledge and time

A worker with knowledge z asks a solver $s > z$ for help.

$$h n(z)(1 - z) = 1 \implies n(z) = \frac{1}{h(1 - z)}.$$

Firm profit

$$\Pi_{HH} = n(z)[s - w(z)] - w(s).$$

Humans maximize income; firms choose teams to maximize profit.

Free entry and market clearing determine wages and AI rents.

More knowledgeable workers need less help, expanding the solver's span.

Conditions: people, problems, and firms

- Unit human mass; one unit of time per human.
- Observable $z \in [0, 1]$ with continuous, positive density.
- Independent uniform problem difficulty; risk neutrality.
- Competition, complete information, output price one.
- No fixed costs; at most two organizational layers.

Conditions: scarce time and abundant compute

- $0 < h < h_0$: no independent humans before AI.
- $a < 1$: AI is not perfectly knowledgeable.
- Production opportunities exceed production capacity.
- Compare autonomy at the same capability a and compute μ .

A sufficient compute-abundance condition

$$\mu > \int_0^a h(1-z) dG(z) + \frac{1 - G(a)}{h(1-a)}.$$

Proposition 5: capability determines bottom gains

$$B = \{z \in [0, a] : w^A(z) > w(z)\}, \quad T = \{z \in [a, 1] : w^A(z) > w(z)\}.$$

Bottom: a capability threshold

$$B \neq \emptyset \iff a > \bar{a}, \quad \bar{a} \in \text{int } W.$$

W is the pre-AI worker set.

- Top: $T \neq \emptyset$ for every $a < 1$, under the stated conditions.
- Winners occupy endpoint intervals; the human with $z = a$ loses.
- Because $\bar{a} \in \text{int } W$, even some basic AI benefits the bottom.

Proposition 6: adoption still requires capability

Co-pilot-only equilibrium has zero AI rent: $r^N = 0$.

Capability	Outcome
$a \leq w(0)$	No adoption; wages and occupations unchanged.
$a > w(0)$	Lowest-knowledge humans adopt; strict bottom gains and some strict human losses.

Capability condition: $a > w(0)$ is required for strict bottom gains.

Proposition 6: local comparisons, explicit quantifiers

Near the bottom

$$\exists \epsilon_B > 0 \ \forall z \in [0, \epsilon_B) : \quad w^N(z) \geq \max\{w(z), w^A(z)\}.$$

Both comparisons are strict if $a > w(0)$.

Near the top

$$\exists \epsilon_T > 0 \ \forall z \in (1 - \epsilon_T, 1] : \quad w^N(z) \leq w^A(z).$$

Strict for $z \neq 1$; no strict ranking is asserted at $z = 1$.

w : before AI; w^A : autonomous AI; w^N : co-pilot-only AI.

Proposition 6: equilibrium, losses, and output

- The equilibrium is unique and efficient; it maximizes labor income within the co-pilot technology.
- With adoption, AI-assisted workers occupy the bottom. Human-assisted workers and human solvers also exist; independent humans may be absent.
- Some $z \in (0, 1]$ satisfies $w^N(z) \leq w(z)$; strict if $a > w(0)$.

Output under the same capability and compute supply

$$Y^A > Y^N.$$

Why autonomy changes the distribution

For a human worker assisted by AI, zero profit implies

$$w = a - h(1 - z)r.$$

Autonomous AI: $r^A = a$

AI can produce on its own.

Advice has an opportunity cost:

$$w = a[1 - h(1 - z)].$$

Co-pilot AI: $r^N = 0$

Abundant compute can only advise.

The same team's wage becomes

$$w = a.$$

This compares team formulas; equilibrium can also change human roles.

A three-type economy we can solve directly

Type	z	Mass
Low	0	.55
Middle	$1/2$	.30
High	1	.15

$$h = \frac{1}{2}, \quad \mu = 2.$$

Before AI, zero profit gives

$$u + \frac{1}{2}v = \frac{1}{2},$$

$$u + \frac{1}{2}t = 1,$$

$$v + \frac{1}{4}t = 1.$$

Subtract the first two: $t - v = 1$.

Hence $(u, v, t) = (.2, .6, 1.6)$.

This finite example changes the paper's population assumption.

Autonomous AI: solve one equilibrium branch

For $0 < a < 1/3$, spare compute pins the rent at $r^A = a$.

Human role	Wage
High: supervise AI workers	$t = 2$
Middle: supervise AI workers	$v = (1 - 2a)/(1 - a)$
Low: work with middle solvers	$u = \frac{1}{2} - \frac{1}{2}v = \frac{a}{2(1 - a)}$

The .55 low workers use .275 middle solvers.

The remaining .025 middle humans supervise AI and pin v .

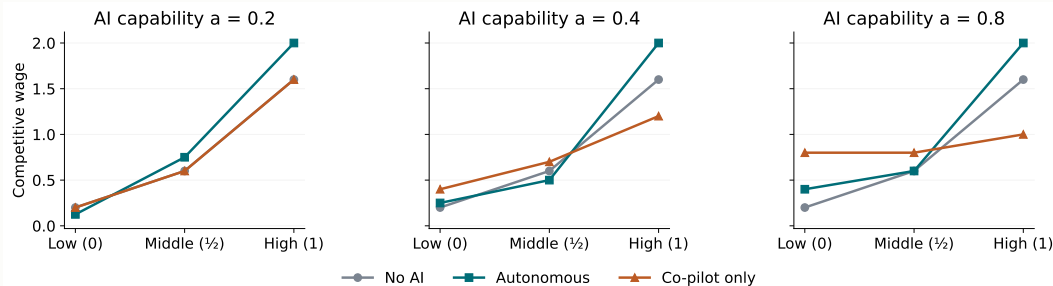
Same autonomy, different bottom outcomes

$$\frac{a}{2(1-a)} > \frac{1}{5} \iff 5a > 2(1-a) \iff a > \frac{2}{7}.$$

Capability	Bottom wage	Versus .20
$a = .20$	.125	Loss
$a = .30$	3/14	Gain

The denominator is positive on $0 < a < 1/3$. Equality is not a gain.

Numerical results separate the two dimensions



Across panels: change capability.

Within each panel: change autonomy.

More output does not imply more labor income

Same capability $a = .4$ and compute supply $\mu = 2$.

	Labor income	Compute income	Output
No AI	.5300	0	.5300
Co-pilot only	.6100	0	.6100
Autonomous	.5875	.8000	1.3875

The numerical program checks feasibility, no profitable entry, primal–dual equality, and wage uniqueness.

What the computational check establishes

Allocation problem

$$\max_{x \geq 0} q^T x \quad \text{s.t.} \quad Ax \leq b.$$

Choose teams subject to human-time and compute constraints.

Price problem

$$\min_{p \geq 0} b^T p \quad \text{s.t.} \quad A^T p \geq q.$$

Find wages and rents that rule out profitable entry.

- Equal objective values certify the finite allocation and prices.
- Wage ranges across optimal prices check uniqueness.

Hand check: capability changes who gains

Total output is: $Y_0 = \frac{K}{2} + Y + Z = \frac{53}{100}$ $\omega_0, v_0, t_0 = (\frac{1}{5}, \frac{2}{5}, \frac{18}{5})$

With autonomous AI: $0 < a < 1/3$

$a - r_A = 0 \rightarrow r_A = a \rightarrow a + \frac{1-a}{2} w_A(s) = s \rightarrow w_A(s) = \frac{2(s-a)}{1-a}$

$\rightarrow t_A = 2, v_A = \frac{1-2a}{1-a}; u_A = \frac{a}{2(1-a)}$

Prop 5 comparison:

Top: $t_A = 2 > 8/5 = t_0$; Bottom: comparing the wage with $\omega_0 = 1/5$

$$\frac{a}{2(1-a)} > \frac{1}{5} \Leftrightarrow a > \frac{2}{7}$$

Claim challenged: "Distribution depends on autonomy, not capability."

$$\frac{a}{2(1-a)} > \frac{1}{5} \Leftrightarrow a > \frac{2}{7}.$$

Here $0 < a < 1/3$, so the denominator is positive.

Verdict: even at fixed autonomy, capability decides bottom gains. This finite threshold is not the paper's $\bar{a}$.

Hand check: wages are not total output

Comparison with non-autonomous AI at the same $a = 3/10$:

$r_N = 0$; $u_N = a = 3/10$; $u_N + t_N/2 = 1 \rightarrow t_N = 7/5$
 $u_N + t_N/4 = 1 \rightarrow v_N = 13/20$

Final comparison:

At $3/10$:	Before AI	Autonomous	Non-autonomous
u	$1/5$	$3/14$	$3/10$
v	$3/5$	$4/7$	$13/20$
t	$8/5$	2	$7/5$

$u_N > u_A > u_0$; $t_A > t_0 > t_N$

Hence, $Y_A > Y_N > Y_0$

At $a = 3/10$, the handwritten wage comparisons are correct.

The last “Hence” needs an extra step: count production, including independent AI.

$$Y_0 = 53/100,$$

$$Y_N = 57/100,$$

$$Y_A = 333/280.$$

Verdict: $Y_A > Y_N > Y_0$. Resource allocations validate this ordering; wages alone do not.

What the paper and the example teach us

- **Bottom:** capability must cross a threshold.
- **Top:** autonomy scales expertise under the stated conditions.
- **Output:** autonomous AI permits additional productive uses.

Next: what can Lean verify?

Six slides on statements, proofs, assumptions, and remaining gaps.

Lean 1/6: from economics to a precise statement

Paper claim $\longrightarrow$ Lean statement $\longrightarrow$ Proof

- A Spec : Prop states objects, assumptions, and conclusion.
- Lean checks a proof of that exact formal statement.
- Matching the statement to the paper requires a separate audit.

Our translation boundary

Real-valued algebra and a finite wage formula are formalized.
The continuum equilibrium and its matching structure are not.

Lean 2/6: the bottom-gain threshold

$$0 < a < \frac{1}{3} \implies \left(\frac{1}{5} < \frac{a}{2(1-a)} \iff \frac{2}{7} < a \right).$$

```
def BottomGainSpec : Prop :=  
  ∀ a : ℝ, 0 < a → a < 1 / 3 →  
    (1 / 5 < bottomWage a ↔ 2 / 7 < a)  
  
theorem bottomGain : BottomGainSpec := by  
  intro a ha hupper  
  have hden : 0 < 2 * (1 - a) := by linarith  
  unfold bottomWage  
  rw [lt_div_iff0 hden]  
  constructor <;> intro h <;> linarith
```

`bottomWage a` means $a/[2(1-a)]$; the domain is $0 < a < 1/3$.

Build: PASS. This is the finite wage inequality, not Proposition 5.

Lean 3/6: why the inequality proof works

```
intro a ha hupper
have hden : 0 < 2 * (1 - a) := by linarith
unfold bottomWage
rw [lt_div_iff0 hden]
constructor <;> intro h <;> linarith
```

- hden: prove $2(1 - a) > 0$ from the stated domain.
- lt_div_iff0: clear division without reversing the inequality.
- constructor: split “if and only if” into two directions.
linarith closes each direction by linear arithmetic.

Identities need nonzero denominators; this inequality needs **positivity**.

Lean 4/6: paper equation, contract, and checked proof

$$n(z) = \frac{1}{h(1-z)}, \quad n(z)(a-w) - r = 0 \implies w = a - h(1-z)r.$$

```
def ZeroProfitWageSpec : Prop :=  
  ∀ h z a w r : ℝ, 0 < h → h < 1 → 0 ≤ z → z < 1 →  
    topAutomatedProfit h z a w r = 0 → w = a - h * (1 - z) * r
```

```
have hhne : h ≠ 0 := ne_of_gt hh  
have hzne : 1 - z ≠ 0 := ne_of_gt (sub_pos.mpr hz)  
unfold topAutomatedProfit span at hprofit  
field_simp [hhne, hzne] at hprofit  
nlinarith
```

h : helping cost; z : worker knowledge; a : AI capability; w : wage; r : rent.

`field_simp` clears nonzero denominators; `nlinarith` finishes.

zeroProfitWage: PASS. Arbitrary real a, w, r broaden the source domain; zero profit is assumed and equilibrium is not proved.

Lean 5/6: a missing assumption changes the claim

Dropping $z < 1$ allows

$$h = \frac{1}{2}, \quad z = 1, \quad a = \frac{1}{2}, \quad w = 1, \quad r = 0.$$

- Lean's real division is total: here $n(z) = 0$.
- Profit is zero, but the proposed wage identity says $1 = 1/2$.
- The separate counterexample compiles successfully.

This tests unrestricted algebra; it is not an admissible economic team.

Lean 6/6: what passed and what remains open

Status	Evidence
PASS	Support proofs, full build, and required fast check.
STOPPED	Semantic preflight: missing source statement map.
0 of 6	Full continuum propositions proved.

Matching, equilibrium, sorting, and source-level audits remain open.

What I learned

A checked proof is useful when its assumptions and scope are explicit.